

Advance research through feedback. Improve strategies through results.

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Abstract

EAR (Effective Auto Research) is an open-source research automation system for idea discovery, autonomous mathematical research, and author responses to peer review. It aims to reduce researchers' active involvement while meeting research quality requirements within a machine budget. Research often advances when plans change: a new paper alters the choice of problem, a counterexample redirects a proof, or a review sends a manuscript back for revision. Existing systems have brought automation to ideation, experimentation, and writing. Sustained research also requires materials to carry forward, feedback to guide the next action, and experience from one task to inform the next.

EAR follows one principle: **use feedback to advance research, and research outcomes to improve its strategies**. AutoVibeIdea combines literature retrieval, critique, and tree search to turn a research direction into an executable proposal. AutonomousMath uses an inner loop for derivation, validation, and manuscript revision, and an outer loop to test and evolve execution strategies through complete research episodes. AutoRebuttal starts from reviewer concerns and available evidence, fills gaps, and compares and revises responses. The three modules can be used independently; researchers can also carry proposals, manuscripts, and review materials between stages. People retain control of direction, key technical checks, and final review.^[1]

In historical evaluations, sustained revision increased current-version model-assessment passes from 3 to 12 among the same 28 manuscripts. A selected evolved strategy produced more passes than the initial strategy while requiring 52.8% fewer research invocations per pass. These findings provide initial support for EAR's approach to sustained research and strategy improvement.^[10]

Keywords Research automation; active human time; critical tree search; strategy evolution; evidence-based rebuttal

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Reading guide Section 1 introduces EAR's scope and central principle. Section 2 explains the three workflows, with particular attention to the mathematical research loops. Section 3 presents manuscript revision, strategy comparisons, and rating-change prediction, together with their evaluation conditions.

1 Introduction

1.1 EAR: keep research moving

Many research decisions arise when a plan changes. A newly discovered paper can remove the reason to pursue a promising idea. A counterexample can force a proof onto a different route. A gap identified in review can send a nearly finished manuscript back to derivation or experiment. Research advances through these specific turns. An automation system needs to follow them: preserve the work already done, understand what the new evidence changes, and continue.

EAR (Effective Auto Research) is an open-source research automation system for idea discovery, autonomous mathematical research, and author responses to peer review. Once a researcher supplies a direction and constraints, retrieval, investigation, and revision should be able to continue. When the researcher returns, there should be new work to inspect and clear reasons for deciding which directions merit further effort.

EAR offers three independently usable workflows. Idea discovery starts from a direction and resource constraints, compares literature and candidates, and develops a research proposal. Mathematical research pursues a specific target and delivers arguments, implementations, and manuscripts. Rebuttal uses the paper, reviews, and available evidence to organize responses and retain their supporting sources. Researchers can use these materials to connect work across stages.^[1]

Sustained work also needs better ways of doing research. In the mathematics module, EAR puts both aims into a pair of loops: the inner loop carries out the current research; the outer loop tests execution strategies against complete research outcomes. Target selection, proof organization, and revision instructions can therefore improve with experience. **Research outcomes and the strategies that produce them are the two objects EAR seeks to improve.** The system works on the present task and uses success, failure, and review to inform its next attempt.

Reserve limited human time for the key judgments

Active involvement in research is spread across reading, coordination, checking, and rework. Reconstructing context after an interruption adds another cost.^[6,7] EAR therefore places active human time at the center of its design objective:

$$\min H \quad \text{subject to} \quad Q \geq Q_{\min}, \quad C \leq B. \quad (1)$$

Here H is active human time, $Q_{\min}$ is the required research quality, C is the cost of models, tools, and computation, and B is the machine budget. Calendar duration T is recorded separately and may impose a deadline constraint. For collaborative work, H sums the actual time contributed by all participants.

This objective requires accounting for the full workload. Literature search and reading, task coordination, technical checks, repairs to model errors, rework on failed approaches, and final review all count toward human time. Additional oversight must count too. Unattended waiting belongs to calendar duration; the time spent waiting without active work does not enter H . This

report presents process records, model assessments, and invocation counts. Actual savings in active human time remain to be measured directly.

1.2 The challenge: turn feedback into the next research step

Existing work has introduced automation into several parts of research. ResearchAgent develops ideas from the literature and iterates on them through review feedback. The AI Scientist connects ideas, experiments, result analysis, and writing. Paper2Rebuttal organizes evidence and response plans around reviewer concerns. GEPA uses execution traces and textual feedback to evolve prompts. These systems provide useful foundations for sustained research.^[2,3,4,5]

In practice, researchers first face the problem of carrying materials forward. A proposal's assumptions, closest related work, and validation plan remain useful when derivation begins. A failed proof should leave behind its counterexample and the point of failure. When these judgments are scattered across interactions, the next person must reread and reorganize them. What research state is saved, how it is handed over, and when human intervention is required all affect the burden of continued use.

Carrying feedback forward requires another step. A low score does not specify the next action: the gap may concern the proof, the contribution, or its presentation. The system needs to translate review comments into the appropriate research work, repairing a derivation, supplying missing evidence, or bringing the manuscript's claims back into line with the established results.

Finally, experience needs to inform the next research episode. A plausible new instruction still has to survive target selection, technical work, writing, and evaluation before its value is known. Changes to research strategy should be judged by complete task outcomes. Table 1 summarizes the objects addressed by related work.

EAR brings these three forms of continuity under one objective: saved research materials support further execution, feedback leads to specific changes, and complete outcomes determine subsequent strategies. For researchers repeatedly screening directions, pursuing mathematical or algorithmic tasks, or addressing reviewer concerns, this organization provides a basis for continuing work and inspecting consequential decisions.

System	Research or optimization object	Role of feedback
ResearchAgent ^[2]	Literature-based questions, methods, and experimental ideas	Review comments guide idea revision
The AI Scientist ^[3]	Machine learning ideas, experiments, results, and papers	Experimental results support iteration; automated review informs manuscript evaluation and the knowledge archive
GEPA ^[4]	Prompts in language model systems	Execution traces and reflection guide prompt mutation and combination
Paper2Rebuttal ^[5]	Reviewer concerns, evidence, and response plans	Produces an inspectable plan before organizing the response
EAR ^[1,10]	Research tasks and their execution strategies	Feedback advances the current task; the mathematical outer loop tests strategies through complete research

Table 1: Research objects and feedback mechanisms. This is a comparison of design scope; performance on shared tasks has not been tested. The AI Scientist refers to the original 2024 system and excludes the later AI Scientist-v2.

1.3 EAR’s principle: feedback for research, results for strategy

In the mathematics module, EAR uses a complete research episode as the unit for evaluating strategy evolution. The inner loop retrieves, derives, validates, and revises toward the current target. The outer loop runs each candidate strategy through research, then uses manuscripts, assessments, and failure reasons to decide what to retain or change. Problems exposed by one episode can inform the next episode’s target requirements, proof organization, and revision instructions. **Each research episode should leave a better starting point for the next.**

The same principle informs the other two workflows. Idea discovery uses new literature to reassess candidate value: supported proposals receive further refinement, while overlapping or infeasible directions retain their exit reasons. Rebuttal locates the evidence relevant to a concern, fills the gaps, and organizes the answer. Feedback can thus redirect effort and lead to concrete technical work.

Human judgment runs through all three workflows. Researchers set direction and quality standards, check key technical conclusions, and take responsibility for final manuscripts and responses. EAR continues retrieval, investigation, and revision, preserving the reasons behind candidate selection, version changes, and strategy updates. Researchers should be able to resume from work that has advanced, with the evidence for their next judgment.

From a design principle to an observable result Among the same 28 completed manuscripts from G0–G3, current-version model-assessment passes rose from 3 to 12 after three revisions. Section 3.2 reports the round-by-round trajectory and retained-version results. Passes here follow the historical model-assessment rule.^[10]

2 System design and methods

This section explains EAR’s architecture and three workflows. Section 3 examines manuscript revision, strategy comparisons, and rating-change prediction.

EAR’s three workflows advance from saved research state: existing materials enter the next round of execution, and new evidence changes what follows. Mathematical research adds strategy evolution, using complete research outcomes to revise the way tasks are carried out.

2.1 System architecture: preserve the state of research

An execution starts from a direction, resources, and delivery requirements. It uses tools for retrieval, assistance with derivation, implementation, and evaluation, and saves the research materials and feedback. The next round continues from that work. Figure 1 shows the relationship between proposals, manuscripts, and review materials. The three modules accept inputs independently; researchers organize handoffs between stages. The evidence in this report comes from separate module records. A complete cross-module workflow has not yet been evaluated separately.

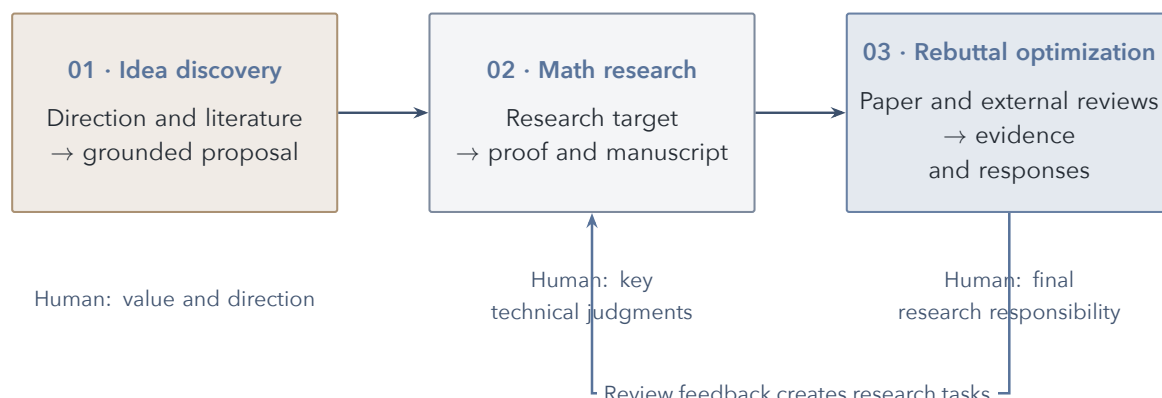


Figure 1: EAR’s research tasks and artifact handoffs. Each workflow is independently usable; proposals, manuscripts, and review materials can supply later stages. People retain key judgments and responsibility for the research.

Workflow	Inputs and outputs	Human judgments
Idea discovery	Direction and constraints → literature comparison, candidates, and proposal	Problem value and execution route
Mathematical research	Direction or proposal → arguments, code, manuscript, and assessment	Key technical conclusions and acceptance of the work
Rebuttal	Paper, reviews, and evidence → response and evidence ledger	Scope of claims and final responsibility

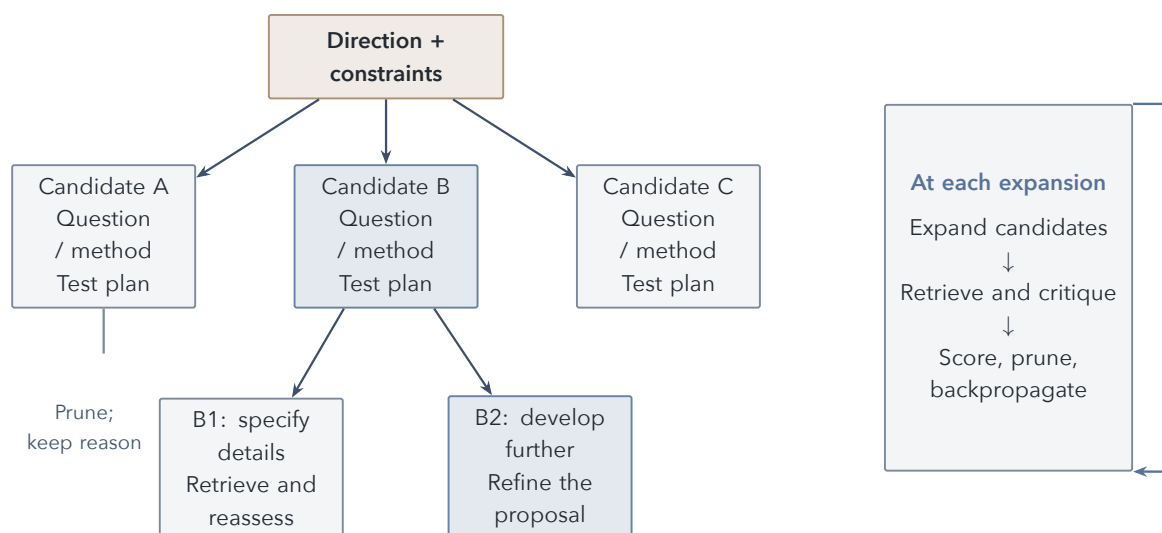
Table 2: Inputs, deliverables, and human responsibilities. Researchers organize stage handoffs through proposals, manuscripts, and review materials.

Hand over the judgments already made

When someone takes over research midway, the most useful handoff is a record of decisions: which assumptions were adopted, which papers are closest, how far the proof has progressed, and which routes have failed. EAR saves this state and the remaining validation plan so work can continue from existing results. Evaluation records also distinguish actual tool feedback from the host’s self-review, allowing researchers to inspect important choices.^[1]

2.2 Idea discovery: make each candidate more specific

Idea discovery starts from a research direction and resource constraints, then develops candidates into specific questions, assumptions, methods, and validation plans. Each expansion introduces fresh retrieval and critique. Evidence updates node value and changes where effort goes next. Figure 2 shows candidate expansion and branch termination.



UCT-guided best-first expansion; retrieved evidence updates which branches merit further work.

Figure 2: Candidate expansion and critical search. Retrieval and evaluation update node value. Overlapping, infeasible, or duplicate branches can terminate, with their reasons preserved.

2.2.1 Retrieve and critique at every expansion

The current implementation uses UCT to guide best-first expansion. The model proposes child nodes; evaluation tools update visit counts and node values and backpropagate along the parent chain. Each new candidate is checked again for at least novelty and research fit. Node selection combines scores, verifiable literature overlap, validation rules, and structural signals. The search evaluates expanded candidates directly, without random rollouts to simulate terminal outcomes.^[1,2]

Table 3 lists the four evaluation fields and their weights. Scores, reasons, and retrieval sources are saved together, making the decision to expand a branch traceable. Different agents

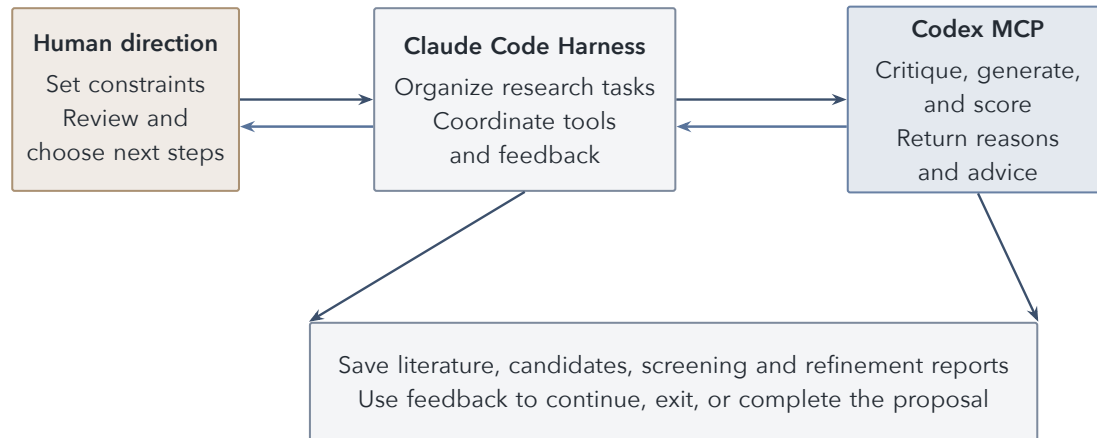
perform generation, critique, and evaluation. The extent to which reviewer personas are used is determined by the actual prompts and execution records.

Field	Research question examined	Weight
Novelty	Does the core mechanism differ substantively from recent work?	.25
Venue	Do the contribution and rigor meet the target requirements?	.35
Strategic	Is the claim falsifiable, is evidence available, and does the plan fit the resources?	.20
Feasibility	Are implementation complexity, computation, data, and timing feasible?	.20

Table 3: Current search evaluation fields and weights. Assessments guide the next expansion; sources and reasons support later checks.

2.2.2 From research intent to an executable proposal

The researcher supplies a direction, resource constraints, and key questions. Claude Code Harness organizes tasks and schedules tools; Codex MCP returns candidates, critique, scores, and suggestions. Figure 3 shows one interactive integration of the ideation module. Feedback returns to the current task, and screening and refinement outputs are saved as inputs to the next step.



Direction and constraints Candidate screening Proposal refinement
Interactive integration: humans advance stages; saved outputs feed the next decision.

Figure 3: Interactive integration of idea discovery. The researcher sets constraints and chooses the next step. The harness and evaluation tools exchange feedback and save candidates, literature, and refined proposals.

Candidate screening and proposal refinement can be performed separately. During screening, the evaluator reads the candidate’s central claim and closest literature, assesses novelty and substantive differences, and recommends continuing or abandoning it. Refinement develops retained candidates by specifying the research setting and validation plan. The early integra-

tion advances through staged interactions; the current tree search further organizes candidate expansion. Both preserve the materials needed to continue work.^[1]

2.2.3 A rejected direction still leaves evidence

Search can terminate a branch when a candidate substantially overlaps existing work or is infeasible under the stated conditions. EAR retains the node, sources, difference analysis, and exit reason so later work can use that negative evidence. Retained candidates are refined around a specific problem, with explicit assumptions, main arguments, implementation interfaces, and a minimum validation plan, producing an executable proposal.

Process case E1: close prior work changes the decision

In an early staged reassessment, additional retrieval found three closely related works across two candidates. Candidate A's novelty score fell from 8/10 to 4/10, and the recommendation changed from continuation to abandonment. Candidate B fell from 8/10 to 5/10. New literature drove the change: the system reassessed the core differences and revised the scores, reasons, and recommended investment.^[10]

The initial retrieval had not adequately covered close prior work, making the candidates appear more distinctive. Additional retrieval changed that judgment and the subsequent action. This record shows how rescreening can affect investment decisions. It comes from the early staged workflow, which differs from the current UCT search; performance comparisons between algorithms still require suitable controls.

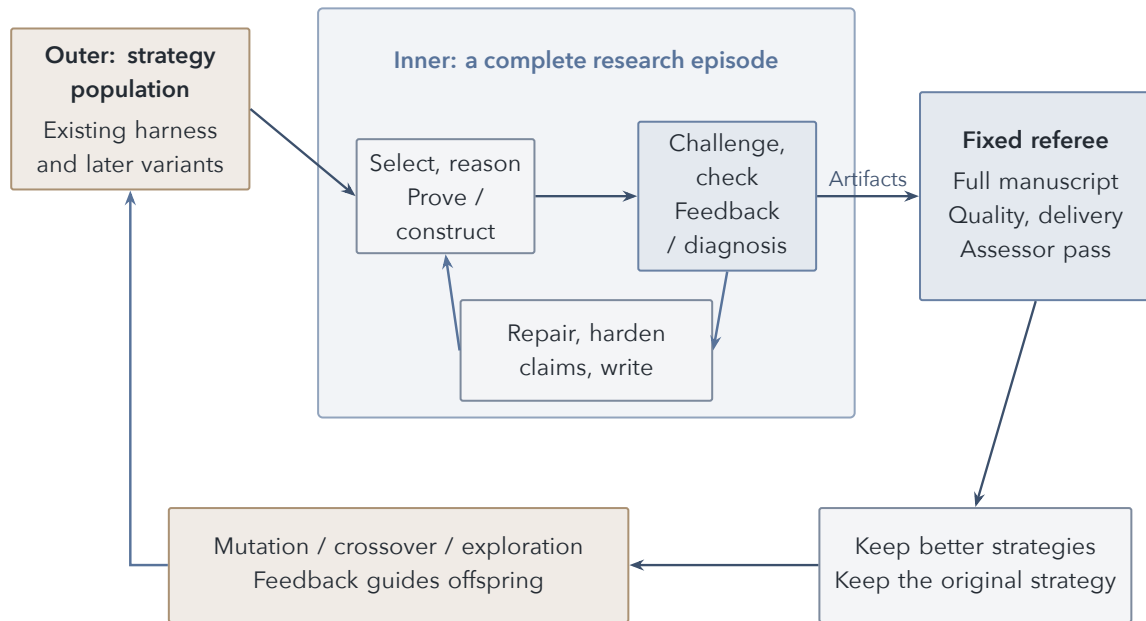
2.3 Mathematical research: solve the task, improve the strategy

Mathematical and algorithmic research can turn technical concerns into specific propositions, counterexamples, constructions, and numerical checks. EAR follows these leads in the inner loop, then passes the outcome of a complete research episode to the outer loop to test the execution strategy. Figure 4 shows the nesting of research revision and strategy evolution.

2.3.1 Save research state; revise the specific gaps

The inner loop carries out target selection, reasoning, proof or construction, implementation, checking, and writing toward a defined goal. Precise propositions, failed proof steps, counterexamples, numerical validation, and experimental controls all enter the research state. After an attempt, the next step can pursue an unresolved gap. The system completes a manuscript when delivery conditions are met and records an exit when it cannot continue.

A proof gap sends revision back to derivation. Implementation or validation problems trigger further technical checks. An insufficient contribution requires reconsidering scope and claims; presentation problems call for rewriting. Historical runs performed three revisions and reassessments after the initial review, then restored the manuscript version with the highest mean score. Every revision and assessment change is retained. Current-round trajectories and final delivered versions are therefore reported separately, following Section 3.1.



Each candidate runs the full research procedure; historical strategies seek their own targets.

Figure 4: The mathematical research loops. The inner loop advances the current task; the outer loop compares and updates strategies using complete outcomes while retaining the initial strategy. Historical strategies could select their own specific targets.

2.3.2 Test every strategy through a complete research episode

An outer-loop genome represents a candidate research strategy. In the historical version, it consists of stage prompts and strategy parameters for target screening, proof organization, writing, and revision. The current portable version expands the mutable scope to task programs, prompts, skills, tool workflows, and configuration, while keeping base model weights fixed. Initial and candidate strategies are saved separately; lineage, research outcomes, and failure reasons inform later selection.^[1,4]

A strategy earns adoption through the results of the research it performs. Every candidate must attempt complete research. Target selection, technical progress, manuscripts, assessments, and failure exits all enter outer-loop selection. In the current portable version, the task list, final referee, and standards remain fixed outside the candidate’s mutable scope. Historical experiments allowed strategies to choose specific research targets. Section 3.1 interprets the results under that historical setting.

Four operations update strategies. **Elitism** retains stronger strategies. **Directed mutation** changes a parent in response to exit reasons, technical gaps, or low-score comments. **Semantic crossover** combines complementary instructions from two parents. **Exploration** tries larger changes. Historical changes addressed target choice and argument development: shortening candidate lists, raising target and contribution requirements, and requesting core lemmas and feasible proof routes earlier. The updated strategies then execute research again and are judged by their results.

2.3.3 Success and exit both count toward fitness

Historical episode fitness is computed as

$$F(e) = \begin{cases} q_{\text{best}}(e), & \text{completed manuscript with a valid score,} \\ 0.5, & \text{research exit,} \\ 0, & \text{screened out,} \\ -0.5, & \text{no target or search failure,} \\ -2, & \text{budget exhausted or contribution too trivial.} \end{cases} \quad (2)$$

Here q_{best} is the highest manuscript mean score retained across the initial assessment and revisions. If a manuscript is completed but all scoring attempts fail, the historical implementation uses a fallback of 3.0. Research exit, screening rejection, search failure, and budget exhaustion have distinct fitness values, so decisions and failures before successful writing also enter strategy evaluation.

Strategies in a generation are ranked and selected by mean F . The historical elite archive separately records performance through

$$A_{\text{hist}} = \frac{1}{3} \left(\frac{\bar{F} + 2}{12} + w + a \right), \quad (3)$$

where w is the manuscript completion rate and a the historical model-assessment pass rate. The current portable version equally weights normalized quality, completion rate, and fixed final-referee pass rate:

$$A_{\text{portable}} = \frac{1}{3} \left(\frac{\bar{q}_{\text{clip}}}{10} + w + a_{\text{final}} \right). \quad (4)$$

Current quality scores are clipped and normalized under the specified rules. When combined scores are comparable, invocation counts and rework are compared next. Mean F serves historical within-generation selection, A_{hist} serves the historical archive, and A_{portable} serves the current version. Invocation efficiency is calculated separately from execution records.^[1,10]

2.4 Rebuttal: resolve the concern before writing the response

Rebuttal starts from the paper, reviewer comments, available evidence, and deadline. It first checks whether the relevant proof or experiment already exists, then arranges necessary additional work, drafts a response, and retains its supporting evidence. Figure 5 follows the path from concern to evidence to answer.

2.4.1 Find the evidence behind each answer

Several reviewers may raise the same concern from different angles. EAR groups related comments, locates the supporting argument in the paper, and distinguishes missing explanation from missing proof or experiment. It organizes existing evidence when the explanation is inadequate and schedules additional research when a technical gap remains. One proof or experiment can answer several related comments. An evidence ledger records claims and their sources.^[1,5,9]

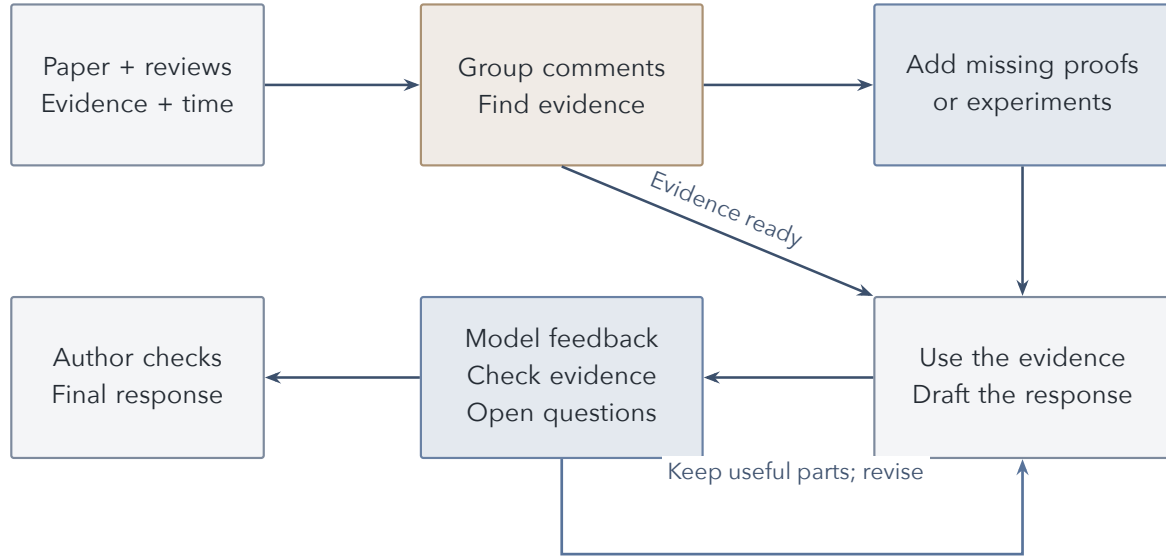


Figure 5: Locate concerns and evidence in the paper and reviews. Draft directly when evidence is sufficient; fill technical gaps first when it is missing. Model feedback guides revision, and authors check key content and the final response.

Public community methods and author experience supply the initial strategies. Candidates differ in answer order, evidence choice, and organization, prioritizing concerns that affect the assessment of the paper. The objective under resource constraints is

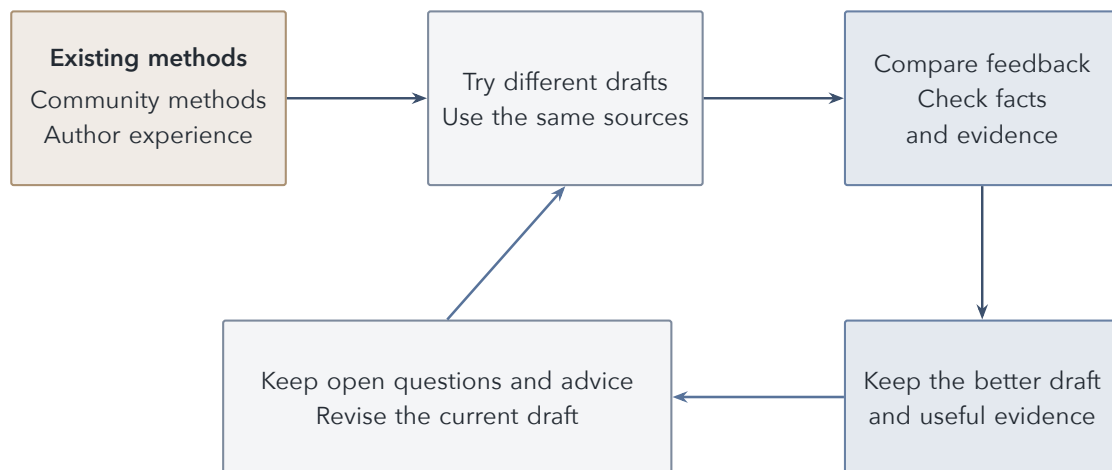
$$\max_{\theta} \mathbb{E}[\Delta R \mid \text{Context}, \theta] \quad \text{subject to} \quad H \leq H_B, C \leq B, T \leq D. \quad (5)$$

Here Context contains the paper, reviews, and available evidence; θ is the response strategy; ΔR is the desired rating improvement; H_B is the human-time limit; and D is the deadline. This equation states a design objective. The prediction tests examine the feedback component, while process records examine strategy adjustment. The effect of generated responses on real reviewer ratings has not been measured directly.

2.4.2 Keep the useful draft; track what remains unresolved

Candidate strategies may lead with existing evidence, organize arguments around decisive concerns, or fill missing validation first. They use the same paper and review materials and compare both the work plan and the response approach. Figure 6 shows how drafts and feedback enter the next round.

Useful drafts are retained, along with unresolved concerns, evidence gaps, and revision advice. This is black-box optimization: the system compares candidate responses and adjusts strategies from diagnostic feedback, without computing a derivative of the rating with respect to the strategy. Evaluation checks concern coverage and evidential support and predicts whether ratings rise, remain unchanged, or fall. Section 3.4 separately tests this prediction capability.



Compare drafts, keep useful work, revise, and compare again.

Figure 6: Initialize response strategies from community methods and author experience. Candidates share the same materials. Model evaluation and fact checks provide feedback; the next round inherits useful drafts, evidence, unanswered questions, and revision advice.

Process case: two feedback rounds change the strategy

One anonymous historical record compares three strategies in each round: leading with existing evidence, organizing around decisive concerns, and adding evidence in stages. The evidence-first strategy selected in round one did not meet the historical internal proxy's stopping requirement. Round two inherited the draft and revision advice; selecting the concern-first strategy then triggered the stop condition. This record shows how feedback changed the approach. It has not been reassessed under current thresholds, and real reviewer rating changes were not measured.^[10]

The historical workflow stops trying the remaining strategies in a round once a draft passes content evaluation. It then checks fidelity to the paper and evidence and the accuracy of the wording. Problems return the draft to revision; the final response goes to the author for review. Authors confirm new technical conclusions, evidence validity, and the final response.

3 Evaluation and results

The evaluation examines two kinds of improvement: how manuscripts change through sustained revision, and what results research strategies achieve after evolution. Rating-change prediction is tested separately to assess whether the feedback component can identify changes following existing author responses.

3.1 Evaluation protocol: attempts, manuscripts, and passes

After inspecting the archive, we selected G0 initialization and three generations of evolution, G1–G3, as the reporting window. The archive also contains later generations. This window records 64 research attempts, 630 research invocations, 28 completed manuscripts, and 15

model-assessment passes for the best retained versions. Failed attempts and attempts without completed manuscripts remain in the totals.^[10]

Unit or definition	Meaning in this report
Research attempt (episode)	A complete research execution; failures enter totals and invocation counts
Research invocation	A launch of the research-stage executor recorded by the historical driver
Completed manuscript	The required manuscript artifacts were produced; completion is counted separately from passing review
Current-round pass	The version under review in that round meets the historical Accept rule
Best retained-version pass	Uses the assessment labels of the restored version with the highest mean score
Fixed inner-loop subset	The same 28 completed manuscripts, tracked through initial assessment and three revisions
Strategy comparison batch	Four research attempts for each initial or selected strategy in a generation

Table 4: Units and version definitions in the mathematical evaluation. Completion, current-round passes, and retained-version passes answer distinct questions.

Historical assessment and version restoration

Each historical assessment round starts three fresh Codex sessions to evaluate a manuscript using a simulated SAC scale. At least one Accept vote counts as a pass. Initial assessment is followed by three rounds of revision and reassessment. The version with the highest mean score is then restored, along with its assessment labels. A pass here refers to the historical model-assessment rule. Real conference acceptance and the current portable version’s fixed final review are separate outcomes.

Manuscript revision tracks only the 28 completed manuscripts. Failed and unfinished attempts remain in the outer-loop statistics. Each generation’s initial and selected strategy batches contain four attempts each. In G0 and G1, the two roles refer to the same batch, which is counted only once in the overall ledger.

Strategy comparison and invocation coverage

Historical strategies select specific targets from a supplied direction, so comparisons reflect both target choice and execution. The batches are adaptive, unpaired historical runs. They did not preselect identical propositions, impose equal token, dollar, or human-time budgets, or include ablations of evolutionary operators. We report counts for the selected strategies, full populations, and entire window separately, making both strategy adoption and strategy search visible.

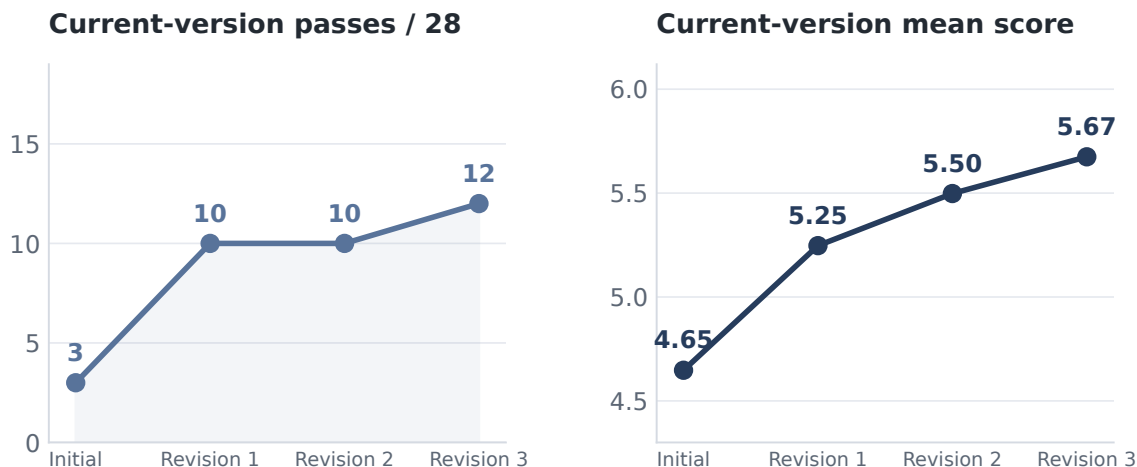
The invocation ledger covers research-stage launches, independent review sessions, and revision attempts inferred from assessment rounds after the initial one. Each review group contains three independent sessions. Totals count sessions; group counts are not added again. The combined invocation count is a cost proxy. It does not cover strategy coaching and breeding, internal model and tool calls, unrecorded retries, full billed tokens, dollars, or active human time.

Process evidence and prediction metrics

Idea-screening case E1 comes from early staged reassessment. The two-round rebuttal record uses a historical internal proxy's stopping rule. These records show changes in execution decisions. Rating prediction instead uses existing papers, reviews, and responses, with labels given by the sign of final minus initial ratings. The mathematical statistics can be checked against public aggregate data; prediction metrics retain the source report's values. The evaluation protocol accompanies the report.

3.2 Manuscript revision: current passes rise from 3 to 12

The same 28 completed manuscripts undergo initial assessment and three revisions. Current-version passes are 3, 10, 10, and 12, while mean scores rise from about 4.65 to 5.67. Figure 7 shows the current-version trajectory. Table 5 also reports the best versions retained through each round.



G0-G3: 28 completed manuscripts. Best-version retention yields 15 final passes separately.

Figure 7: The fixed subset of 28 manuscripts from G0-G3. Current-version passes and mean scores are shown for initial assessment and three revisions. Best retained-version results appear in Table 5.

Round	Current passes	Current mean	Retained passes	Retained mean
Initial	3	4.648	3	4.648
Revision 1	10	5.247	10	5.270
Revision 2	10	5.498	12	5.568
Revision 3	12	5.675	15	5.790

Table 5: Round-by-round results for the fixed manuscript subset, with 28 manuscripts in every round. Means are recomputed from recorded score sums and displayed to three decimal places; full aggregate values are included in the released data. Retained columns refer to the best versions through each round.

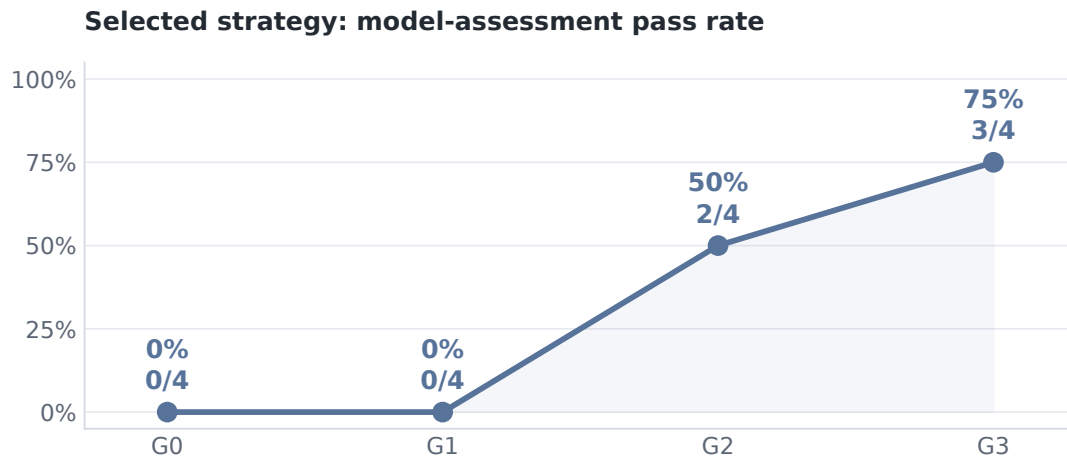
The current-version pass proportion increases from $3/28 = 10.7\%$ to $12/28 = 42.9\%$, with an initial-to-final mean-score gain of about 1.03. Relative to initial assessment, 27 manuscripts improve in mean score at the last round and one declines. Model assessments of this subset improve overall after revision. The records lack a reassessment-only control, however, so they do not isolate the causal benefit of feedback-guided revision.^[10]

Keep the highest-scoring version when revision regresses

Version trajectories also include regressions. Thirteen manuscripts have lower means in the last round than at their own historical best. Three have a passing best version but a non-passing last version. Restoring the highest-mean versions yields 15 retained passes, compared with 12 current passes at the final round. These regressions illustrate the practical role of version history: revision can continue while an earlier, stronger manuscript remains available for delivery.

3.3 Selected G3 strategy: more passes, fewer invocations

Figure 8 reports the historical pass rate of each generation’s selected strategy. Each point represents four complete research attempts, including failed and unfinished attempts in the denominator. The selected batches in G0 and G1 have no passes; G2 has 2/4 and G3 has 3/4. Each generation uses its own selected strategy and research targets.



G0 initialization, then three generations. Each point covers four attempts with that generation's selected strategy.

Figure 8: Selected-strategy pass rates for G0 initialization and G1–G3 are 0%, 0%, 50%, and 75%. Each point represents four historical research attempts, rather than repeated measurement of one strategy on fixed tasks.

Turn the last episode's problems into the next strategy's changes

The outer loop brings exit reasons and review comments into strategy revision. Recorded changes include shortening candidate lists, requesting core lemmas and proof routes earlier, and raising contribution requirements. Subsequent batches test those changes through research outcomes. Table 6 gives the initial and selected strategy counts; Table 7 gives the full population ledger.

Gen.	Initial passes / attempts	Selected passes / attempts	Initial research invocations	Selected research invocations
G0	0/4	0/4	53	53
G1	0/4	0/4	40	40
G2	0/4	2/4	24	45
G3	2/4	3/4	41	29

Table 6: Historical batch comparison of the initial strategy and each generation's selected strategy. In G0 and G1, both roles refer to the same batch and cannot be counted twice in the full population.

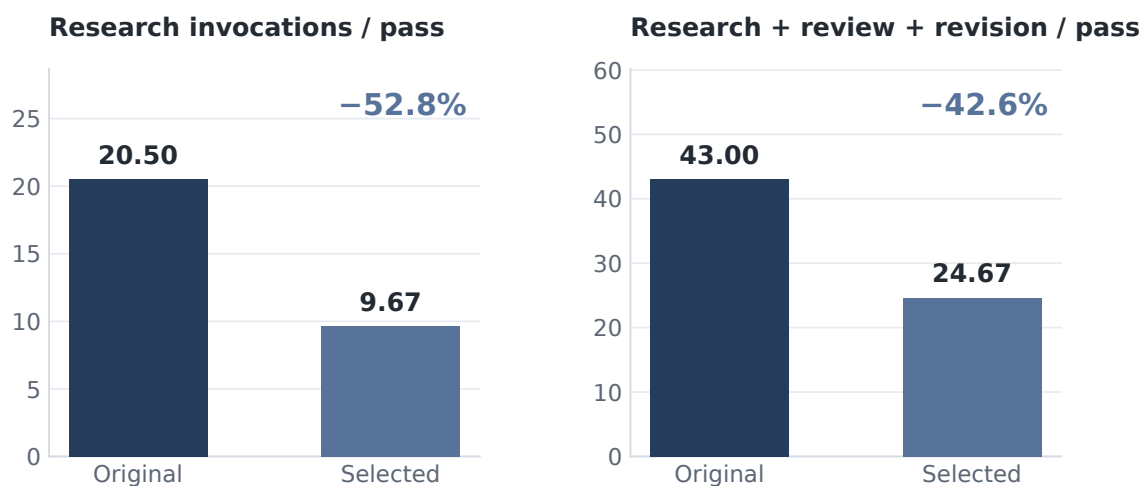
Gen.	Attempts	Research invocations	Completed papers	Retained passes	Pass rate
G0	12	169	1	0	0.0%
G1	16	155	5	1	6.25%
G2	16	148	8	5	31.25%
G3	20	158	14	9	45.0%
Total	64	630	28	15	23.4%

Table 7: Full G0–G3 population counts. Failed and unfinished attempts remain in the denominator. Selected batches and population-wide performance are reported separately; retained passes use the best-version rule.

The G3 selected strategy passes 75% of attempts ($3/4$), compared with 50% ($2/4$) for the initial strategy. Counting all candidates and the initial strategy gives a G3 population rate of 45% ($9/20$) and an overall window rate of $15/64 = 23.4\%$. Restricting the denominator to completed manuscripts gives $15/28 = 53.6\%$. These scopes distinguish adopting a selected strategy, conducting strategy search, and assessing completed manuscripts.

Research invocations per pass fall by 52.8%

Both G3 strategy batches perform four research attempts. The initial strategy uses 41 research invocations and obtains two passes; the selected strategy uses 29 and obtains three. Figure 9 and Table 8 put research, review, and inferred revision in a common ledger.



G3: four attempts per strategy. Historical model assessment; revision counts reconstructed from records.

Figure 9: In G3, research invocations per pass fall from $41/2 = 20.50$ to $29/3 = 9.67$, a 52.8% reduction. The combined research, review, and inferred-revision proxy falls from $86/2 = 43.00$ to $74/3 = 24.67$, a 42.6% reduction.

G3 strategy	Research	Review	Revision*	Total	Passes	Total / pass
Initial	41	36	9	86	2	43.00
Selected	29	36	9	74	3	24.67

Table 8: G3 invocation ledger, with four attempts per batch. Each batch has 12 review groups and 36 independent sessions. Revision* is inferred from assessment rounds after the initial one; review groups are not added again to the session total.

Research invocations per attempt fall from $41/4 = 10.25$ to $29/4 = 7.25$, while passes increase from two to three. Together these produce the 52.8% reduction in research invocations per pass. Including review and inferred revision gives a 42.6% reduction in the combined proxy per pass. Section 3.1 defines both scopes; full machine costs and human time are not included.

Counting the entire search process, G0–G3 use 630 research invocations for 15 passes, or 42 research invocations per pass. This ledger includes the investment in finding strategies and is reported separately from the cost of adopting the G3 selected batch. Invocations per pass are undefined for batches with zero passes.

The pass definition changes the comparison

We also apply a stricter diagnostic to the retained review votes: all three votes must be Accept, and each score must be at least 7. Both the G3 initial and selected strategies satisfy this requirement in only 1/4 attempts. No new reviews were run. This diagnostic shows how the historical pass rule affects the interpretation of the advantage. Small batches, self-selected targets, and a reporting window chosen after inspecting the archive also limit generalization. Stable benefits require matched tasks, consistent quality checks, and full resource accounting.^[10]

3.4 Rating prediction: stronger on rises and ties, weaker on falls

The rebuttal feedback evaluator reads a paper, its initial review, and the author’s recorded response, then predicts whether the rating rises, stays unchanged, or falls. Each case is supplied separately. The final rating is used only to check the label, with actual change given by $\text{sign}(R_{\text{final}} - R_{\text{initial}})$.

The test joins public OpenReview historical records with ProReviewer initial ratings. It uses DeepSeek V4 Pro, isolated cases, zero-shot prompting, and temperature 0. Table 9 reports aggregate results from the source report. Macro-F1 averages the F1 scores across classes; higher values, up to 1, are better.^[1,8,10]

Source and sample	Prediction setting	Macro-F1
36-case balanced subset	18 rises and 18 unchanged ratings; excludes falls	0.805
57-case full test	Rise, no change, and fall; includes six falls	0.503
Same 57-case source set	Source-reported rise/no-change metric*	0.755

Table 9: Prediction of post-response rating changes. *The third row comes from the same 57-case source set, but its effective binary denominator was not reported separately. The rows use different evaluation settings. F1 values are retained from the source; public materials do not include per-case predictions for reestimation.

The balanced binary test yields a Macro-F1 of 0.805; the full three-class test yields 0.503, with F1 of 0 for the falling-rating class. Detecting falls is a clear weakness of this evaluator, which the binary test does not cover. Further prompt changes did not improve held-out performance, so the original zero-shot prompt was retained. Model weights were not changed.

EAR combines rating-change predictions with evidence checks to compare drafts. The current test asks whether rating changes can be predicted following existing responses. Whether newly generated responses improve real reviewer ratings requires a separate experiment. Response decisions still require evidence, content checks, and author judgment, particularly given the evaluator’s weak detection of falling ratings.

Public materials and result verification

Released counts, score sums, and the evaluation protocol allow readers to check means, proportions, invocation totals, and cost per pass. Anonymous process records preserve specific turns where evidence changed a decision or feedback changed a strategy. The public aggregates do not include individual research targets, complete manuscripts, per-vote review text, per-case rebuttal predictions, or full execution logs. They support statistical checks; reproducing the complete research process requires those additional materials.

4 Conclusion

The next research step is often determined by evidence just obtained. EAR implements this mode of progress in three workflows. Idea discovery reassesses candidates against close prior work. The mathematical inner loop revises research around gaps identified in review, while the outer loop tests execution strategies through complete outcomes. Rebuttal finds evidence relevant to reviewer concerns, retains useful drafts, and continues filling gaps. Research materials, decision reasons, and version changes are saved along the way, providing the basis for further work.

Historical records offer initial evidence for these mechanisms. The fixed manuscript subset receives better model assessments after revision. In its historical batch, the G3 selected strategy achieves more passes and fewer research invocations per pass. Rating-change prediction shows both binary discrimination and a weakness in detecting falls. Each result has a specific task, denominator, and evaluation condition. The cost of using an existing strategy also needs to be distinguished from the cost of continuing to search for strategies.

Keep research moving; keep people on the key decisions

Researchers set the direction and standards and check key results. EAR continues retrieval, investigation, and revision within those constraints, returning new work and the evidence needed to decide what follows.

The ultimate test is the human time required to produce trustworthy research. Future evaluation should use matched tasks and consistent quality checks, recording time spent on reading, coordination, checking, repairs, failed-route rework, and final review. It should also account for full machine costs and calendar duration. These measurements are needed to establish how far EAR's design objective is achieved in practical use.

The open-source project and public materials

The [EAR open-source repository](#) provides idea discovery, autonomous mathematical research, and rebuttal modules. Aggregate data, the evaluation protocol, and vector figures accompany this report to support statistical and interpretive checks; Section 3.4 describes their scope and limitations. Readers can try the relevant workflows, check the reported results, and contribute further improvements.^[1,10]

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